SOFTWARE ENGINEERING LAB

LAB 5

STATIC CODE ANALYSIS

Hostel Room Allocation System

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1. Introduction

This report documents the static code analysis and code-quality improvement work performed for Lab 5 on the Hostel Room Allocation System. The objective was to inspect the Python source file hostel_allocation.py with Pylint, Bandit and Flake8, identify quality, security and style issues, correct them, and verify the updated program. In addition to the issues flagged by the tools, the program output was read carefully, because several logic errors (such as exceeding room capacity) are not detected by any of the three tools.

2. Objectives

- Analyse the Python program using Pylint, Bandit and Flake8.
- Identify code-quality, security and formatting issues reported by the tools.
- Find logic errors by reading the program output, and correct them while preserving the program's purpose.
- Re-run all three tools and record the final results.
- Document the issues, fixes, execution evidence and reflections.

3. Tools and Environment

Tool / Component	Purpose / Version
Python	Python 3.13.7 (virtual environment)
Pylint 4.1.3	Code quality and static analysis (with Astroid 4.3.4)
Bandit 1.9.4	Python security issue detection
Flake8 7.5.0	Style and formatting checks (pycodestyle 2.15.0, pyflakes 4.1.0)
Flask 3.1.3 / PyYAML 6.0.3	Libraries used by the application
OS / Shell	Windows, PowerShell

4. Application Overview

The Hostel Room Allocation System reads a YAML room configuration (Block A: male, 3 rooms, capacity 2, fee 4000; Block B: female, 2 rooms, capacity 3, fee 3500) and provides functions to allocate beds by block and gender, vacate rooms, calculate fees in 30-day periods, order students by seniority, run a room lottery, count vacant beds, simulate a warden database connection, and look up a student's room through a small Flask web page. The source file analysed is hostel_allocation.py.

5. Initial Static Analysis

The original source was analysed before making any change. The baseline results showed problems in code quality, security and formatting.

Tool	Initial result	Interpretation
Pylint	5.84/10	32 findings: 28 convention and 4 warning messages (naming, missing docstrings, unused imports, dangerous default value, unused argument, line length)
Bandit	5 issues	1 High (B201 Flask debug=True), 2 Medium (B506 unsafe yaml.load, B612 logging.config.listen), 2 Low (B311 random, B106 hardcoded password)
Flake8	20 violations	13 × E302, 3 × E501, 2 × F401, 1 × E721, 1 × E305

5.1 Baseline Evidence

```
(venv) PS C:\Users\LAPTOPS24\Downloads\L5_60_hostel-room-allocation> python hostel_allocation.py
Rooms: ['A1', 'A2', 'A3', 'B1', 'B2']
S1 -> A1
S2 -> A1
S3 -> A1
S4 (female) -> A2
S1 again -> A2
S5 year '2' -> None
Occupants of A1: ['S1', 'S2', 'S3']
Occupants of A1 after S2 left: ['S1', 'S2', 'S3']
Fee for S1, 45 days: 4000
Priority (seniors first?): [1, 2, 4]
Lottery: ['S6', 'S7', 'S8']
Vacant beds: 8
Warden DB: {'host': 'localhost', 'user': 'warden', 'connected': True}
Waitlist: []
```

Figure 1 : Original program execution

Reading the original output shows several logic errors that none of the tools report: S3 was placed in A1 although its capacity is 2; the female student S4 was placed in A2, a male block; "S1 again" created a second allocation in A2; S2 was still listed in A1 after leaving; the priority order was [1, 2, 4] (juniors first); and thevacant-bed count o F8 was wrong.

```
(venv) PS C:\Users\LAPTOPS24\Downloads\L5_60_hostel-room-allocation> Get-Content .\pylint_report_before.txt
************* Module hostel_allocation
hostel_allocation.py:95:0: C0301: Line too long (116/100) (line-too-long)
hostel_allocation.py:1:0: C0114: Missing module docstring (missing-module-docstring)
hostel_allocation.py:20:0: C0116: Missing function or method docstring (missing-function-docstring)
hostel_allocation.py:20:0: C0103: Function name "loadConfig" doesn't conform to snake_case naming style (invalid-
ame)
hostel_allocation.py:29:0: C0116: Missing function or method docstring (missing-function-docstring)
hostel_allocation.py:29:0: C0103: Function name "allocateRoom" doesn't conform to snake_case naming style (invali
-name)
hostel_allocation.py:29:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
hostel_allocation.py:30:7: C0123: Use isinstance() rather than type() for a typecheck. (unidiomatic-typecheck)
hostel_allocation.py:29:35: W0613: Unused argument 'gender' (unused-argument)
hostel_allocation.py:41:0: C0116: Missing function or method docstring (missing-function-docstring)
hostel_allocation.py:41:0: C0103: Function name "vacateRoom" doesn't conform to snake_case naming style (invalid-
ame)
hostel_allocation.py:47:0: C0116: Missing function or method docstring (missing-function-docstring)
hostel_allocation.py:47:0: C0103: Function name "monthlyFee" doesn't conform to snake_case naming style (invalid-
ame)
hostel_allocation.py:52:0: C0116: Missing function or method docstring (missing-function-docstring)
hostel_allocation.py:52:0: C0103: Function name "priorityOrder" doesn't conform to snake_ca
d-name)
hostel_allocation.py:55:0: C0116: Missing function or method docstring (missing-function-do
```

Power & Battery

- *Figure 2 : Pylint before corrections*

```
(venv) PS C:\Users\LAPTOPS24\Downloads\L5_60_hostel-room-allocation> Get-Content .\bandit_report_before.txt
Run started:2026-10-31 03:09:59.127850+00:00

Test results:
>> Issue: [B506:yaml_load] Use of unsafe yaml load. Allows instantiation of arbitrary objects. Consider yaml.safe_
load().
   Severity: Medium   Confidence: High
   CWE: CWE-20 (https://cwe.mitre.org/data/definitions/20.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b506_yaml_load.html
   Location: .\hostel_allocation.py:21:13
20      def loadConfig(text=CONFIG):
21          config = yaml.load(text, Loader=yaml.Loader)
22          for block in config["blocks"]:

--------------------------------------------------
>> Issue: [B311:blacklist] Standard pseudo-random generators are not suitable for security/cryptographic purposes.
   Severity: Low   Confidence: High
   CWE: CWE-330 (https://cwe.mitre.org/data/definitions/330.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b311-random
   Location: .\hostel_allocation.py:56:11
55      def roomLottery(student_ids):
56          return random.sample(student_ids, len(student_ids))
```

Figure 3(a) : Bandit before corrections

```
--------------------------------------------------
>> Issue: [B106:hardcoded_password_funcarg] Possible hardcoded password: 'Hostel@123'
   Severity: Low   Confidence: Medium
   CWE: CWE-259 (https://cwe.mitre.org/data/definitions/259.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b106_hardcoded_password_funcarg.html
   Location: .\hostel_allocation.py:66:11
65      def connectWardenDb():
66          return sqlite_connect(host="localhost", user="warden", password="Hostel@123")
67

--------------------------------------------------
>> Issue: [B612:logging_config_insecure_listen] Use of insecure logging.config.listen detected.
   Severity: Medium   Confidence: High
   CWE: CWE-94 (https://cwe.mitre.org/data/definitions/94.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b612_logging_config_insecure_listen.html
   Location: .\hostel_allocation.py:72:11
71      def startRemoteLogConfig():
72          return logging.config.listen(9999)
73
```

Figure 3(b).

```
(venv) PS C:\Users\LAPTOPS24\Downloads\L5_60_hostel-room-allocation> Get-Content .\flake8_report_before.txt
hostel_allocation.py:1:1: F401 'os' imported but unused
hostel_allocation.py:2:1: F401 'sys' imported but unused
hostel_allocation.py:20:1: E302 expected 2 blank lines, found 1
hostel_allocation.py:25:80: E501 line too long (92 > 79 characters)
hostel_allocation.py:29:1: E302 expected 2 blank lines, found 1
hostel_allocation.py:30:8: E721 do not compare types, for exact checks use `is` / `is not`, for instance checks us
e `isinstance()`
hostel_allocation.py:41:1: E302 expected 2 blank lines, found 1
hostel_allocation.py:47:1: E302 expected 2 blank lines, found 1
hostel_allocation.py:52:1: E302 expected 2 blank lines, found 1
hostel_allocation.py:55:1: E302 expected 2 blank lines, found 1
hostel_allocation.py:58:1: E302 expected 2 blank lines, found 1
hostel_allocation.py:65:1: E302 expected 2 blank lines, found 1
hostel_allocation.py:66:80: E501 line too long (81 > 79 characters)
hostel_allocation.py:68:1: E302 expected 2 blank lines, found 1
hostel_allocation.py:71:1: E302 expected 2 blank lines, found 1
hostel_allocation.py:74:1: E302 expected 2 blank lines, found 1
hostel_allocation.py:79:1: E302 expected 2 blank lines, found 1
hostel_allocation.py:82:1: E302 expected 2 blank lines, found 1
hostel_allocation.py:95:80: E501 line too long (116 > 79 characters)
hostel_allocation.py:101:1: E305 expected 2 blank lines after class or function definition, found 1
```

Figure 4. Flake8 before corrections (20 violations)

6. Issues Identified and Corrections

6.1 Security issues (Bandit)

Issue	Bandit ID / Severity	Line (original)	Correction applied
Unsafe YAML loading with yaml.Loader	B506 – Medium	21	Replaced with yaml.safe_load() and added a check that the configuration is a mapping containing "blocks".
Insecure random generator for the lottery	B311 – Low	56	Replaced random.sample() with secrets.SystemRandom().shuffle() on a copy of the list.
Hardcoded database password	B106 – Low	66	The password is now read from the WARDEN_DB_PASSWORD environment variable.
Insecure logging.config.listen(9999)	B612 – Medium	72	Removed the remote listener; the function now only logs that it is disabled.
Flask run with debug=True	B201 – High	80	Changed to app.run(debug=False).

6.2 Logic errors found from program output

Issue	Line (original)	Correction applied
Capacity check used ">" so rooms could exceed capacity	34	Rooms are skipped when occupants >= capacity.
Gender not checked; parameter "gender" unused (Pylint W0613)	29, 32–38	A room is allocated only if its block gender matches the student's gender.
Same student could be allocated twice	31–38	The existing allocation is returned if the student ID is already allocated.
Unknown preferred room caused KeyError; no waitlist de-duplication	32–39	Unknown rooms are skipped and a student is added to the waitlist only once.
vacateRoom did not free the bed	41–45	Student is removed from both the allocations dictionary and the room's occupants list; returns True/False.
Fee function did not validate input	47–50	Raises ValueError for negative days or a student without a room.

Issue	Line (original)	Correction applied
Priority order was ascending	53	Sorted by year in descending order so seniors come first.
Vacant beds counted only fully empty rooms	58–63	Computed as capacity minus occupants for every room.
Reflected HTML and crash in /room lookup	76–77	Missing/unknown IDs return 400/404 messages; only the stored room number is shown.

6.3 Code-quality and style issues (Pylint and Flake8)

Issue	Tool / Code	Correction applied
camelCase function names (11 occurrences)	Pylint C0103	Renamed to snake_case (load_config, allocate_room, vacate_room, monthly_fee, priority_order, room_lottery, vacant_beds, connect_warden_db, start_remote_log_config, room_lookup, run_portal).
Missing module and function docstrings	Pylint C0114, C0116	Added a module docstring and a docstring for every function.
Mutable default argument preferred=[]	Pylint W0102	Default changed to None and the list is created inside the function.
type(year) != int comparison	Pylint C0123, Flake8 E721	Replaced with isinstance(); booleans and values below 1 are rejected.
Unused imports os and sys	Pylint W0611, Flake8 F401	Removed sys; os is now used to read the environment variable.
Iterating over dict keys and indexing	Pylint C0206	Iterate with .values() / .items().
Blank lines and long lines	Flake8 E302, E305, E501; Pylint C0301	Added two blank lines between definitions, split long lines (max 79 chars) and put the program entry point under if __name__ == "__main__".

7. Final Static Analysis Results

After the corrections, all three tools were executed again. The final source reached a Pylint score of 10.00/10, Bandit reported no remaining issues, and Flake8 produced no output.

Tool	Initial result	Final result	Status
Pylint	5.84/10 (32 findings)	10.00/10	Passed
Bandit	5 issues (1 High, 2 Medium, 2 Low)	No issues identified (0)	Passed
Flake8	20 violations	No output / 0 violations	Passed

7.1 Final Tool Evidence

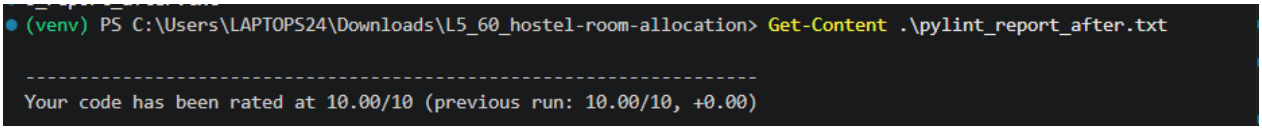


Figure 5. Pylint after corrections (10.00/10)

```
(venv) PS C:\Users\LAPTOPS24\Downloads\L5_60_hostel-room-allocation> Get-Content .\bandit_report_after.txt
Run started:2026-10-31 03:36:48.928042+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 158
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
```

Figure 6. Bandit after corrections (no issues identified)

```
(venv) PS C:\Users\LAPTOPS24\Downloads\L5_60_hostel-room-allocation> python -m flake8 hostel_allocation.py > flake
8_report_after.txt
>> Get-Content .\flake8_report_after.txt
```

Figure 7. Flake8 after corrections (no output)

8. Program Execution and Functional Verification

The corrected hostel_allocation.py was executed again. The program completed successfully and every printed value is now correct:

Output line	Result	Why it is correct
S1 / S2 / S3 ->	A1 / A1 / A2	A1 holds two students (capacity 2); the third male student moves to A2.
S4 (female) ->	B1	A female student is placed only in Block B.
S1 again ->	A1	The existing allocation is returned; no duplicate record.
S5 year '2' ->	None	The year is a string, so the input is rejected.
Occupants of A1 after S2 left	['S1']	The vacated bed is released.
Fee for S1, 45 days	4000	One complete 30-day period × fee 4000.
Priority (seniors first)	[4, 2, 1]	Seniors are listed first.
Vacant beds	9	Total capacity 12 (3 × 2 + 2 × 3) minus 3 allocated students.
Warden DB connected	False	The password is no longer hardcoded; it is empty unless WARDEN_DB_PASSWORD is set.

```
(venv) PS C:\Users\LAPTOPS24\Downloads\L5_60_hostel-room-allocation> python hostel_allocation.py
Rooms: ['A1', 'A2', 'A3', 'B1', 'B2']
S1 -> A1
S2 -> A1
S3 -> A2
S4 (female) -> B1
S1 again -> A1
S5 year '2' -> None
Occupants of A1: ['S1', 'S2']
Occupants of A1 after S2 left: ['S1']
Fee for S1, 45 days: 4000
Priority (seniors first?): [4, 2, 1]
Lottery: ['S7', 'S8', 'S6']
Vacant beds: 9
Warden DB: {'host': 'localhost', 'user': 'warden', 'connected': False}
Waitlist: []
(venv) PS C:\Users\LAPTOPS24\Downloads\L5_60_hostel-room-allocation>
```

Figure 8. Final program execution

9. Submission Structure and Version Control

The Lab 5 submission is stored in the repository L5_60_hostel-room-allocation (SETAPESU26). The folder contains:

- hostel_allocation.py – cleaned source code
- issues.md – issue table
- reflection.md – reflection answers
- pylint_report_before.txt and pylint_report_after.txt
- bandit_report_before.txt and bandit_report_after.txt
- flake8_report_before.txt and flake8_report_after.txt

The work was committed in three Lab 5 commits on the main branch:

Commit	Message
654420b	Fix code quality issues identified by Pylint
b0189a0	Fix security issues identified by Bandit
a4e97d1	Fix formatting issues identified by Flake8

10. Reflection

What is static code analysis and why is it important?

Static code analysis examines source code without running it. It finds defects, unsafe practices and style problems early, which makes the program more reliable, secure and maintainable.

What are the differences between Pylint, Bandit and Flake8?

Pylint evaluates overall code quality: naming, docstrings, unused code and likely programming errors. Bandit looks only for security problems such as unsafe deserialisation, hardcoded passwords and debug mode. Flake8 checks style and formatting (PEP 8) and simple errors such as unused imports. Together they give complementary coverage.

What issues were identified and fixed in this program?

Unsafe YAML loading, an insecure random generator, a hardcoded password, an insecure logging listener, Flask debug mode, camelCase names, missing docstrings, a mutable default argument, a type

comparison, unused imports and formatting violations. From reading the output I also fixed logic errors: rooms exceeding capacity, missing gender restriction, duplicate allocations, vacating not freeing a bed, wrong priority order and a wrong vacant-bed count.

What did you learn from this lab?

Static analysis catches security and style problems before the program is run, but it cannot judge whether the output is correct; the capacity, gender and priority bugs were found only by reading the output. Security, correctness and readability all matter. After the fixes, Pylint scored 10.00/10, Bandit reported no issues and Flake8 reported no violations.

11. Conclusion

The Lab 5 exercise was completed by analysing the Hostel Room Allocation System with Pylint, Bandit and Flake8, correcting the reported and the unreported problems, and verifying the results. The updated program runs correctly, scores 10.00/10 on Pylint, has no Bandit findings and no Flake8 violations. The source code, issue table, reflection and tool reports are included in the repository.